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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

MA2006-1 – LINEAR ALGEBRA, STATISTICAL ANALYSIS & COMPUTING

(For AD)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. If rank of a coefficient matrix and rank of the augmented matrix are equal then system is

- A) equal B) Consistent
C) inconsistent D) None of these

2. The solution of system of equations $x + y + z = 9$, $x - 2y + 3z = 8$, and $2x + y - z = 3$ is

- A) $x = 2$, $y = 3$ and $z = -4$ B) $x = -2$, $y = -3$ and $z = 2$
C) $x = -2$, $y = 3$ and $z = 2$ D) $x = 2$, $y = 3$ and $z = 4$

3. The rank of matrix $\begin{bmatrix} 2 & -1 & 3 & 1 \\ 1 & 4 & -2 & 1 \\ 5 & 2 & 4 & 2 \end{bmatrix}$ is

- A) 3 B) 4
C) 1 D) 2

4. If every minor of order r of a matrix A is zero, then $\rho(A)$ is

- A) equal to r B) greater than r
C) Less than r D) Greater than or equal to r

5. Find the trace of the matrix $A = \begin{bmatrix} 1 & 0 & 6 \\ 0 & 5 & 0 \\ 0 & 4 & 4 \end{bmatrix}$

- A) 0 B) 4
C) 10 D) 1

6. Given $P = \begin{bmatrix} 2 & -1 \\ 5 & 1 \end{bmatrix}$ and $D = \begin{bmatrix} 6 & 0 \\ 0 & -1 \end{bmatrix}$ find A^3 ?

- A) $\begin{bmatrix} 61 & 62 \\ 156 & 154 \end{bmatrix}$ B) $\begin{bmatrix} 61 & 62 \\ 155 & 150 \end{bmatrix}$
C) $\begin{bmatrix} 61 & 60 \\ 155 & 154 \end{bmatrix}$ D) $\begin{bmatrix} 61 & 62 \\ 155 & 154 \end{bmatrix}$

7. If A is diagonalizable then,

- A) $A^n = (PDP^{-1})^n = PD^nP^{-1}$ B) $A^n = (PDP^{-1})^n = PD^nP$
C) $A^n = (PDP^{-1})^n = PD^nP^1$ D) $A^n = (PDP^{-1})^n = PD^nP^n$

8. If $A = \text{diag}[3, -5, 7]$ and $B = \text{diag}[-1, 2, 4]$ then find the value of $2A + 3B$

- A) $\text{diag}[-3, 4, 26]$ B) $\text{diag}[3, -4, 26]$
C) $\text{diag}[3, 4, 26]$ D) $\text{diag}[3, 4, -26]$

9. The characteristic equation of $\begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$ is

- A) $\lambda^3 - 7\lambda^2 + 16\lambda - 10 = 0$ B) $\lambda^3 - 7\lambda^2 + 16\lambda - 12 = 0$
C) $\lambda^3 - 5\lambda^2 + 16\lambda - 12 = 0$ D) $\lambda^3 - 7\lambda^2 + 14\lambda - 12 = 0$



10. The probability of rejecting the null hypothesis, when it is not true is called
 A) Size of a test B) Type I error
 C) Type II error D) Power of a test
11. Student's t-distribution was discovered by
 A) Laplace B) Karl Pearson
 C) W.S. Gosset D) R. A. Fisher
12. The size of the critical region is denoted by
 A) $1 - \beta$ B) $1 - \alpha$
 C) α D) β
13. To test the equality of variances of two populations, we use
 A) t-distribution B) χ^2 -distribution
 C) Gamma distribution D) F-distribution
14. Analysis of variance technique was developed by
 A) W.S. Gosset B) R. A. Fisher
 C) De-Moivre D) Karl Pearson
15. The more suitable sampling procedure for a heterogeneous population is
 A) stratified sampling B) systematic sampling
 C) simple random sampling D) convenience sampling
16. Repetition of the same treatment on different experimental units is called
 A) randomization B) local control
 C) replication D) Repetition
17. The limit of multiple correlation coefficient $R_{1.23}$ is
 A) -1 to 1 B) -1 to 0
 C) 0 to 1 D) -2 to 2
18. Normal distribution is also classified as
 A) Gaussian distribution B) Poisson distribution
 C) Weighted average distribution D) Bernoulli distribution
19. For a trivariate distribution, if the correlation coefficients are $r_{12} = r_{13} = r_{23} = r$, $-1 < r < 1$ then $r_{12.3}$ is
 A) $\frac{(1+r)}{r}$ B) $\frac{(1-r)}{r}$
 C) $\frac{r}{(1-r)}$ D) $\frac{r}{(1+r)}$
20. If r and R denote correlation and multiple correlation coefficient for the data set X_1, X_2, X_3 , which option is correct?
 A) $r_{12} = 0.21, r_{13} = 0.22, r_{23} = 0.23, R_{1.23} = 0.20$ B) $r_{12} = 0.69, r_{13} = 0.22, r_{23} = 0.23, R_{1.23} = 0.69$
 C) $r_{12} = 0.69, r_{13} = 0.22, r_{23} = 0.23, R_{1.23} = 0.21$ D) $r_{12} = 0.24, r_{13} = 0.22, r_{23} = 0.23, R_{1.23} = 0.21$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I		Marks	BT*	CO*	PO*
1. a)	Test for consistency and solve the system of equations by Gauss elimination method $x + y + z = 6$ $x + 2y + 3z = 14$ $x + 4y + 7z = 30$	6	L*2	1	2
b)	Find the eigen values and vectors of the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$	6	L2	2	2
c)	Define i) Trace of a matrix ii) consistency of linear equations	4	L1	1	1

2. a) Define the rank of a matrix. Find the rank of the following matrix using elementary row transformations.

$$\begin{bmatrix} 1 & 3 & 1 & -2 \\ 1 & 4 & 3 & -1 \\ 2 & 3 & -4 & -7 \\ 3 & 8 & 1 & -7 \end{bmatrix}$$

- b) Determine the largest eigen value and corresponding eigen vector of

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Using Rayleigh's power method, take the initial approximation to the eigen vector as $[1 \ 1 \ 1]^T$. Carry out three iterations.

- c) Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -3 \\ -1 & 1 \end{bmatrix}$ verify that $tr(AB) = tr(BA)$

6 L2 1 2

6 L2 2 2

4 L1 2 1

3. a) Solve the system of linear equations using Gauss-Seidel iteration method

$$2x + y - z = 12,$$

$$2x - 8y + 5z = -5$$

$$-5x + 3y + 9z = -17 \quad \text{start with } x^{(0)} = y^{(0)} = z^{(0)} = 0 \text{ and carry out three iterations.}$$

- b) Find the modal matrix and spectral matrix of $A = \begin{bmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{bmatrix}$ and hence find A^5

6 L2 1 1

10 L2 2 2

Unit – II

4. a) For a random sample of 10 persons fed on diet A, the increased weight in pounds in a certain period were 10, 6, 16, 17, 13, 12, 8, 14, 15, 9. For another random sample of 12 persons fed on a diet B, the increased weight in the same period were 7, 13, 22, 15, 12, 14, 18, 8, 21, 23, 10 and 17. Test whether the diets A and B differ significantly as regards their effect on increase in weight. Use 5% level of significance
- b) Five measurements of the output of 2 units have given the following results (in kilograms of material per one hour of operation)

Unit A	14.1	10.1	14.7	13.7	14.0
Unit B	14.0	14.5	13.7	12.7	14.1

Assuming that both samples have been obtained from normal population, test whether the two populations have the same variance. Use 10% level of significance

6 L3 3 2

6 L2 3 2

- c) Write a short note on i) design of experiment ii) replication iii) randomization

4 L1 4 1

5. a) A student obtained ranks in various subjects as follows, Test at 1% level of significance whether there is any difference in ranks in the three subjects.

M_1	7	11	5	4	-
M_2	4	6	8	5	2
M_3	10	8	6	8	-

6 L2 4 2

- b) Write a note on i) simple random sampling ii) systematic sampling iii) stratified sampling iv) cluster sampling

6 L1 4 1

- c) It has been found from an experience that the mean breaking strength particular brand of thread 275.6g with standard deviation of 39.7g. Recently a sample of 36 pieces of thread. Show the mean breaking strength of 253.2g. Can we conclude that the thread has become inferior at the significant level of 5%.

4 L2 3 2

6. a) The following data relates to number of mistakes in each page of book containing 180 pages.

Number of mistakes:	0	1	2	3	4	5 or more
No. of pages:	130	32	15	2	1	0

Test whether Poisson distribution is a good fit to this observed distribution

6 L2 3 2

- b) Data recorded on three varieties of wheat treated with five different fertilizers. The yield per acre is given below.

Varieties of wheat	Fertilizers				
	F1	F2	F3	F4	F5
V1	55	65	60	62	58
V2	70	66	65	68	63
V3	62	52	49	48	52

Use two factor analysis to analyse the data at 5% level of significance

10 L3 4 2

Unit – III

7. a) If $r_{12} = 0.87$, $r_{13} = 0.82$ and $r_{23} = 0.62$ then calculate $r_{12.3}$, $r_{13.2}$ and $r_{23.1}$

6 L2 5 1

- b) Let X be $N_3(\mu, \Sigma)$ with $\mu = \begin{pmatrix} 5 \\ 3 \\ 7 \end{pmatrix}$ and $\Sigma = \begin{pmatrix} 4 & -1 & 0 \\ -1 & 4 & 2 \\ 0 & 2 & 9 \end{pmatrix}$,

Find i) $P(X_1 > 6)$ ii) $P(5X_2 + 4X_3 > 70)$ iii) $P(4X_1 - 3X_2 + 5X_3 < 80)$

6 L3 5 2

- c) Mention any 4 differences between partial and multiple correlation

4 L1 5 1

8. a) Let X and Y be a jointly normal random variable with parameters $\mu_x = 1$, $\sigma_x^2 = 1$, $\mu_y = 0$, $\sigma_y^2 = 4$ and $\rho = \frac{1}{2}$

a) Find $P(2X + Y \leq 3)$

b) Find $COV(X + Y, 2X - Y)$

c) Find $P(Y > 1 | X = 2)$

6 L2 5 2

- b) From the following data obtain $R_{1.23}$, $R_{2.13}$, $R_{3.12}$

X_1	65	72	54	68	55	59	78	58	57	51
X_2	56	58	48	61	50	51	55	48	52	42
X_3	9	11	8	13	10	8	11	10	11	7

10 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
